

Project1_Final_Submission

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1 STAT 5243 Project 1: WallStreetBets Data Pipeline and Exploratory Analytics

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1.1 1. Introduction and Dataset Description

This project analyzes 53,187 posts from `r/wallstreetbets` during the 2020-2021 meme-stock period. The data combines structured engagement variables (`score`, `comms_num`, timestamps, URL type) and unstructured text (`title`, `body`), creating a mixed-modality dataset with real cleaning complexity.

Complexity signals include high structural text missingness (`missing_body_pct` = 53.49%), heavy-tailed engagement behavior, and temporal normalization requirements.

1.2 2. Data Acquisition Methodology

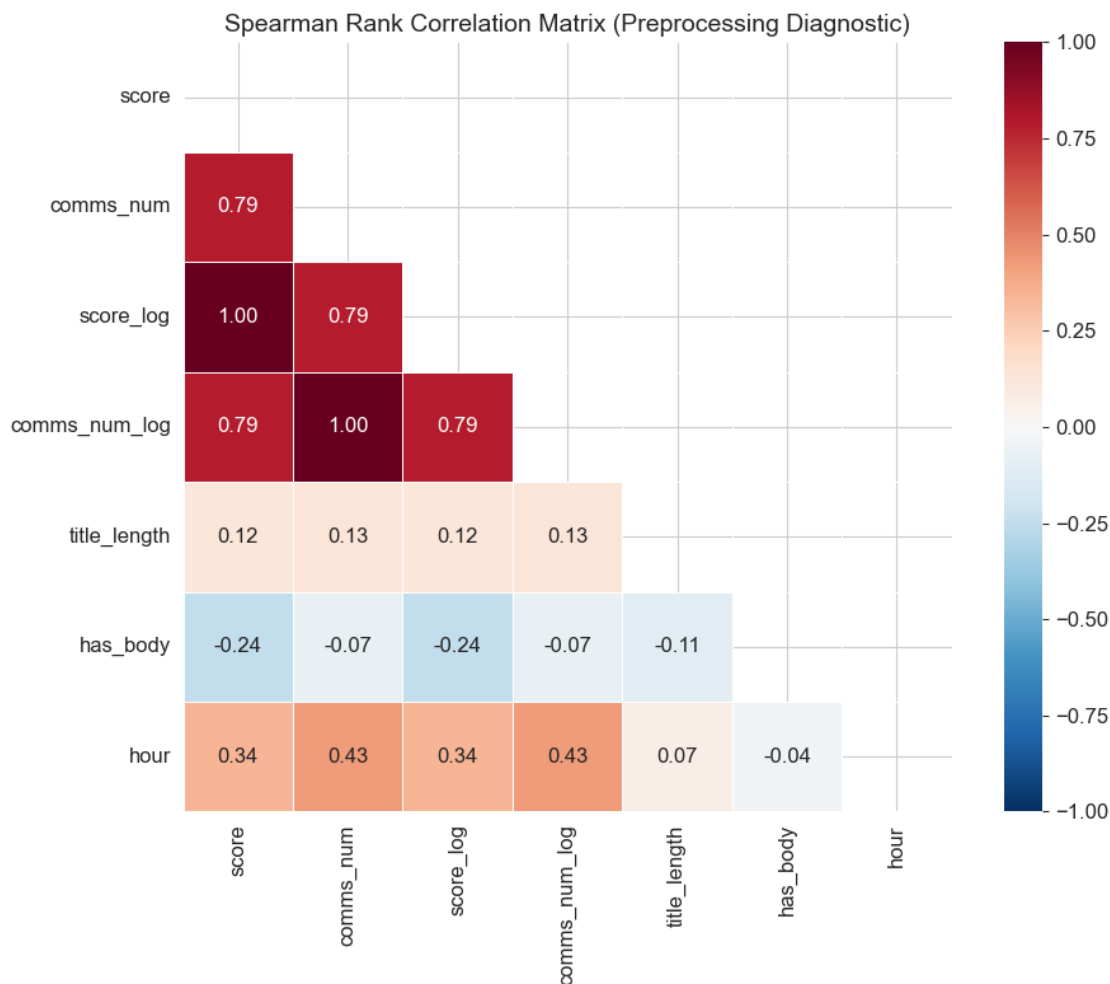
The dataset was obtained from Kaggle (public repository) and originally collected from Reddit via PRAW. We preserved `reddit_wsb.csv` as immutable raw input and produced `reddit_wsb_cleaned.csv` through deterministic preprocessing workflows.

Lineage and integrity checks show row-count preservation (53,187 raw and cleaned), supporting reproducible acquisition-to-processing traceability.

1.3 3. Cleaning and Preprocessing Steps

Cleaning policy highlights: (1) strict datetime coercion (`invalid_timestamp_count_in_raw` = 0), (2) structural missingness handling (`missing_body_pct` = 53.49%) via explicit indicators, (3) duplicate audit (`duplicate id/full-row count` = 0), (4) outlier-aware log transforms, and (5) dual scaling + categorical encoding for model readiness.

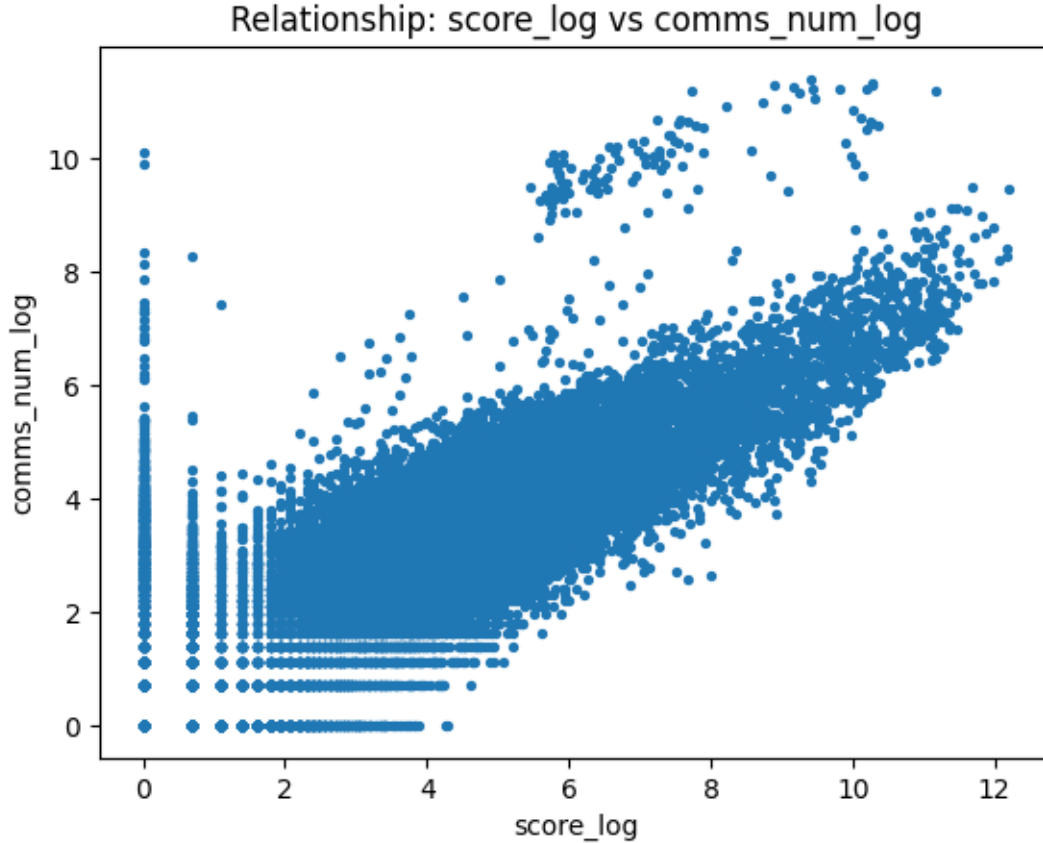
Evidence: - Figure: `.../artifacts/figures/02_cleaning_fig_01_missingness_diagnostics.png`
- JSON: `.../artifacts/json/02_cleaning_policy_metrics.json` -
Section details: `Project Workspace/Supporting Materials/Report Sources/sections/03_cleaning_preprocessing.tex`



1.4 4. Exploratory Data Analysis (EDA)

Advanced EDA quantified engagement dynamics beyond visual summaries. Key results: Spearman correlation between vote and comment activity ($\rho = 0.7852$, $p \ll 0.001$), post-type group difference significance (Kruskal-Wallis statistic 5855.658, $p \ll 0.001$), and viral-tail anomaly rate ($|z| \geq 3$ outlier rate 0.173%).

Interpretation: investor attention is clustered and format-sensitive, with rare but influential extreme events.

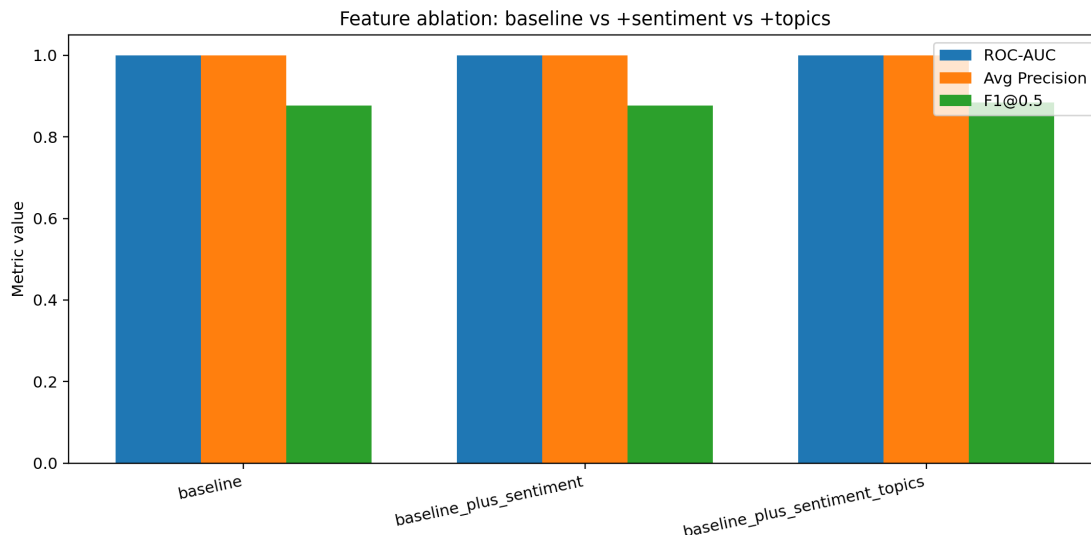


1.5 5. Feature Engineering Process and Justification

Feature engineering added lexicon sentiment and LDA topic probabilities to baseline engagement/temporal predictors, then evaluated deterministic ablation (seed 42).

Model	ROC-AUC	Avg Precision	F1@0.5
Baseline	0.999984	0.999710	0.876950
Baseline + Sentiment	0.999984	0.999710	0.876950
Baseline + Sentiment + Topics	0.999975	0.999546	0.884615

Measured lift: topic features improve thresholded F1 by +0.007666 absolute.



1.6 6. Summary of Key Findings

Key findings: - Structural missingness was high but handled explicitly without dropping representativeness (53.49% body missingness). - Engagement coupling was strong (Spearman $\rho = 0.7852$, $p < 0.001$). - Post-type effects were significant (Kruskal-Wallis $H = 5855.658$, $p < 0.001$). - Topic features improved thresholded viral classification (F1@0.5: 0.876950 $\rightarrow$ 0.884615, +0.007665).

Claim-to-artifact traceability is maintained in Project Workspace/Supporting Materials/Report Sources/main_report.tex and Project Workspace/qa/evidence_registry_report_deptl

1.7 7. Challenges Faced and Future Recommendations

Challenges included structural text missingness, heavy-tailed engagement distributions, and target-definition ease because virality labels are score-derived. Future improvements include forward-looking targets, time-aware validation, and richer NLP (contextual embeddings/sarcasm-aware sentiment), plus external market-data integration for stronger cross-domain inference.

1.8 8. GitHub Repository Link

- <https://github.com/ZemingLiang/STAT-5243-Project-1-Team-2>

1.9 9. Each Member's Contribution

- **Zeming Liang:** Data cleaning and preprocessing; creation/normalization of reddit_wsb.csv, reddit_wsb_cleaned.csv, and Cleaning-and-Preprocessing.ipynb; final repo organization and PNG/JSON/output integration.
- **Zuer Weng:** EDA analysis, advanced statistical diagnostics, and EDA branch artifacts.
- **Duoli Chen:** Report-writing narrative sections (introduction, acquisition, summary, challenges) and narrative refinement.
- **Isaac Beers:** Feature engineering workflow, viral-classification diagnostics, and model interpretation components.